Decision Tree

1. Introduction to Decision Trees

A **Decision Tree** is a supervised machine learning algorithm that makes predictions by learning a sequence of logical decision rules from data.

It works similarly to human decision-making: we ask a series of questions, follow the answers, and eventually reach a decision.

For example, suppose we want to predict whether a student will pass an exam based on their study habits. We might ask:

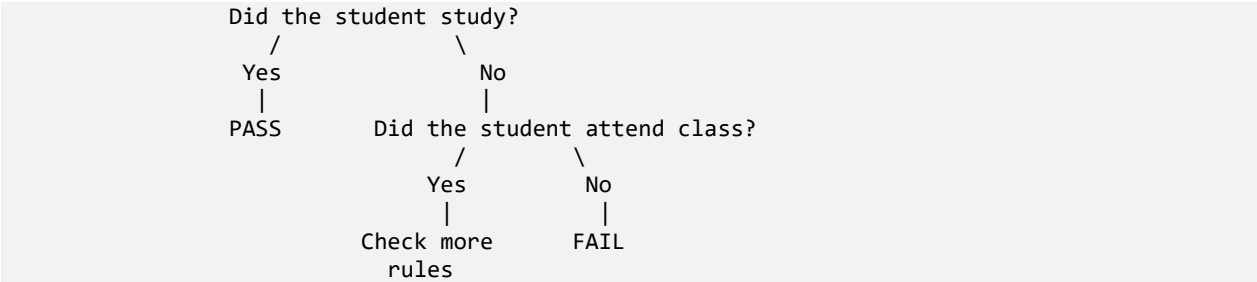
- Did the student study regularly?
- Did the student attend classes?
- How many hours did the student study?

Based on the answers, the model follows a path through a tree and predicts **Pass** or **Fail**.

1.1 How a Decision Tree Works

A decision tree repeatedly divides a dataset into smaller groups using its features. Each split aims to separate the data into groups that are more useful for making predictions.

Consider this simplified example:



Illustrative example only. A real tree learns its decision rules from training data.

1.2 Structure of a Decision Tree

Component	Meaning
Root node	The starting node representing the first decision or split.
Internal node	A node that applies a decision rule to a feature.
Branch	The path representing the outcome of a decision.
Leaf node (terminal node)	The final node that provides the prediction.

1.3 Real-World Applications

Decision trees can be used for tasks such as:

- **Education:** Predicting whether a student will pass or fail.
- **Healthcare:** Classifying patients into risk categories.
- **Banking:** Predicting whether a loan applicant may default.
- **Business:** Predicting customer churn.
- **Manufacturing:** Classifying products as defective or non-defective.

These are examples of possible applications, not results reported by the videos.

2. Types of Decision Tree Tasks

Decision trees can be applied to different supervised learning problems.

Task	Purpose	Example
Classification	Predict a discrete category or class.	Pass/Fail, Spam/Not Spam
Regression	Predict a numerical value.	Predicting a house price

The tree structure is similar in both cases, but the prediction at a leaf differs: a classification tree predicts a class, whereas a regression tree predicts a numerical value.

3. How Does a Decision Tree Choose the Best Split?

A decision tree needs to decide **which feature should be used at each node**.

For example, when predicting whether a student will pass, the available features might be:

- Study hours
- Class attendance
- Assignment completion

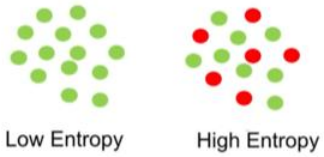
The tree evaluates possible splits and selects one according to a splitting criterion. Two important concepts are **Entropy and Information Gain**. Another is **Gini Impurity**.

These metrics help a decision tree create groups that are more homogeneous in terms of their class labels.

4. Entropy

Entropy measures the uncertainty, randomness, or disorder in a dataset with respect to its class labels.

- High entropy: the classes are mixed, so uncertainty is high.
- Low entropy: one class dominates, so uncertainty is lower.
- Zero entropy: all samples belong to the same class.



For binary classification, entropy is calculated as:

$$\mathbf{H(S) = -p_+ \log_2(p_+) - p_- \log_2(p_-)}$$

Where:

- H(S) = entropy of dataset S
- p_+ = proportion of positive samples
- p_- = proportion of negative samples
- $\log_2$ = logarithm with base 2

For a general classification problem with K classes:

$$\mathbf{H(S) = -\Sigma(i = 1 \text{ to } K)\ p_i \log_2(p_i)}$$

Here, p_i represents the proportion of samples belonging to class i.

If $p_i = 0$, its entropy contribution is defined as zero.

4.2 Example of Entropy

Suppose a dataset contains 10 students:

- 5 students passed.
- 5 students failed.

Therefore,

$$\mathbf{p_+ = (5/10) = 0.5,\ p_- = (5/10) = 0.5}$$

Substituting into the entropy formula:

$$\mathbf{H(S) = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 0.5 + 0.5 = 1 \text{ bit}}$$

The dataset has maximum entropy for a binary classification problem because both classes are equally represented.

Now suppose all 10 students passed:

$$\mathbf{p_+ = 1,\ p_- = 0}$$

Then,

$$\mathbf{H(S) = -1 \log_2(1) - 0 \log_2(0)}$$

$$\mathbf{H(S) = 0}$$

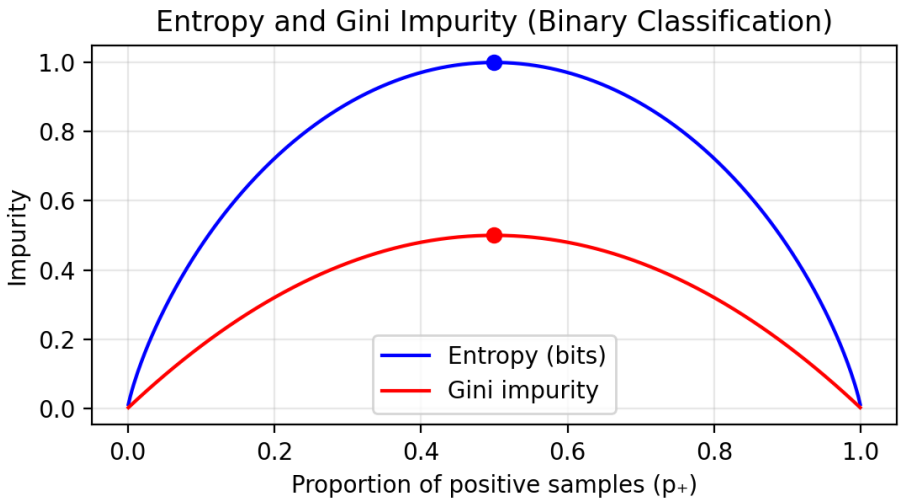
The dataset is pure because every student belongs to the same class.

4.3 Entropy Summary

Class distribution	Entropy
10 passed, 0 failed	0

Class distribution	Entropy
9 passed, 1 failed	Approximately 0.469 bits
5 passed, 5 failed	1 bit

Key takeaway: A decision tree often seeks splits that reduce the entropy of the resulting groups.



5. Information Gain

5.1 Definition

Information Gain measures how much uncertainty decreases after splitting a dataset using a particular feature.

It compares the entropy before a split with the weighted average entropy after the split.

$$IG(S,A) = H(S) - H(S|A)$$

Where:

- IG(S,A) = information gain from splitting dataset S using feature A
- H(S) = entropy before splitting
- H(S|A) = weighted entropy after splitting on feature A

The weighted entropy is:

$$H(S|A) = \Sigma(j = 1 \text{ to } m) (|S_j|)/(|S|) H(S_j)$$

Here, S_j is a group produced by the split, and |S_j|/|S| is its proportion of the original dataset.

Decision rule: When using Information Gain, the tree generally chooses the feature with the highest gain among the candidate splits.

5.2 Worked Example

Suppose the dataset contains 10 students:

- 5 passed
- 5 failed

The initial entropy is:

$$\mathbf{H(S) = 1 \text{ bit}}$$

Suppose splitting by **Study Habit** produces two groups:

Group	Passed	Failed	Entropy
Studies regularly	4	1	0.722
Does not study regularly	1	4	0.722

Each group contains five students. The weighted entropy after splitting is:

$$\mathbf{H(S|A) = (5/10)\ (0.722) + (5/10)\ (0.722) = 0.722}$$

Therefore,

$$\mathbf{IG(S,A) = 1 - 0.722 = 0.278 \text{ bits}}$$

The split reduces uncertainty by approximately 0.278 bits.

Now imagine another feature separates the students into two pure groups:

- Group 1: 5 passed, 0 failed.
- Group 2: 0 passed, 5 failed.

Both groups have zero entropy. Thus,

$$\mathbf{IG(S,A) = 1 - 0 = 1 \text{ bit}}$$

This split perfectly separates the classes in this illustrative dataset.

Conclusion: The second split has higher Information Gain and would be preferred over the first, assuming both are valid candidate splits.

6. Gini Impurity

6.1 Definition

Gini Impurity measures the probability of incorrectly classifying a randomly selected sample if its label is assigned according to the class distribution within a node.

For K classes:

$$\mathbf{Gini(S) = 1 - \Sigma(i = 1 \text{ to } K)\ p_i^2}$$

Where p_i is the proportion of samples belonging to class i.

For binary classification:

$$\mathbf{Gini(S) = 1 - p_+^2 - p_-^2}$$

Interpretation:

- Gini impurity of 0: the node is pure.
- Higher Gini impurity: the node contains a greater mixture of classes.
- For two equally represented classes, Gini impurity is 0.5, its maximum for binary classification.

6.2 Example

Consider the same dataset of 10 students:

- 5 passed
- 5 failed

Therefore,

$$\mathbf{p_+ = 0.5, p_- = 0.5}$$

$$\mathbf{Gini(S) = 1 - (0.5)^2 - (0.5)^2 = 1 - 0.25 - 0.25 = 0.5}$$

Now suppose all 10 students passed:

$$\mathbf{p_+ = 1, p_- = 0}$$

$$\mathbf{Gini(S) = 1 - 1^2 - 0^2 = 0}$$

The node is pure.

6.3 Gini Impurity After a Split

Like entropy, Gini impurity can be evaluated before and after splitting. The tree usually selects a split that minimizes the **weighted Gini impurity** of its child nodes.

$$\mathbf{Gini_split = \Sigma(j = 1 \text{ to } m)\ (|S_j|)/(|S|)\ Gini(S_j)}$$

The reduction in impurity can be expressed as:

$$\mathbf{\Delta Gini = Gini(S) - Gini_split}$$

A larger reduction means the split has improved class separation more strongly.



7. Entropy vs Information Gain vs Gini Impurity

Concept	What it measures	Preferred result when selecting a split
Entropy	Uncertainty or disorder in a node	Lower entropy
Information Gain	Reduction in entropy after a split	Higher information gain
Gini Impurity	Impurity of the class distribution	Lower weighted child impurity, or larger reduction

An important distinction: **Entropy and Gini Impurity measure node impurity, whereas Information Gain measures the reduction in entropy caused by a split.**

Entropy and Gini Impurity often produce similar decisions, but they may rank candidate splits differently.

8. Advantages and Disadvantages of Decision Trees

Advantages

1. **Easy to understand:** Their rules can be visualized and explained.
2. **Interpretable:** It is often possible to trace why a prediction was made. Not sensitive to outliers.
3. **Flexible:** Decision trees can perform classification and regression.
4. **Feature-based decisions:** They select features to split the data according to a chosen criterion.

Disadvantages

1. **Overfitting:** A deep tree may learn noise or specific details in the training dataset rather than general patterns.
2. **Limited generalization:** A tree that overfits may perform poorly on unseen data.
3. **Sensitivity to data:** Small changes in training data can sometimes produce a substantially different tree.
4. **Greedy splitting:** Choosing the best split at each step does not necessarily produce the globally optimal tree.

What Is Overfitting?

Overfitting occurs when a model learns the training data too closely, including noise and accidental patterns. As a result, it performs well on training data but poorly on new, unseen data.

Common ways to control decision-tree overfitting include:

- Limiting the maximum depth.
- Requiring a minimum number of samples for a split or leaf.
- Pruning unnecessary branches.
- Evaluating the model using validation data or cross-validation.

9. Quick Revision Sheet

Concept	Formula	Remember
Entropy	$H(S) = -\Sigma(i)\ p_i \log_2 p_i$	Measures uncertainty; lower is purer.
Information Gain	$IG = H(S) - H(S\|A)$	Measures entropy reduction; higher is preferred.
Gini Impurity	$Gini(S) = 1 - \Sigma(i)\ p_i^2$	Measures impurity; lower is purer.

- **Decision Tree:** A supervised learning model that predicts through feature-based decisions.
- **Root node:** Starting point of the tree.
- **Internal node:** Applies a decision rule.
- **Branch:** Represents an outcome of a rule.
- **Leaf node:** Produces the final prediction.
- **Overfitting:** Strong performance on training data but poor generalization to unseen data.

10. Self-Test Questions

Try answering these questions before moving to the next topic.

1. Which node is the starting point of a decision tree?
2. What does entropy measure?
3. What is the entropy of a pure node?
4. What does Information Gain measure?
5. For binary classes with proportions 0.5 and 0.5, what is Gini impurity?
6. What is overfitting?

Answer Key

1. **Root node.** It is the starting point of the tree.
2. **Uncertainty in class labels.** Entropy measures the uncertainty or disorder of the class distribution.
3. **0.** A pure node contains only one class.
4. **The reduction in entropy after a split.** Information Gain equals parent entropy minus weighted child entropy.
5. **0.5.** Gini = $1 - (0.5)^2 - (0.5)^2 = 0.5$.
6. **Learning training details too closely while generalizing poorly.** The model may perform well on training data but poorly on unseen data.